Power Budget Analysis

1. System Architecture & Assumptions

- **Power Source: 3S LiPo Battery (11.1V, 1800mAh) yielding a total capacity of 19.98 Wh.**
- **Regulator Efficiency: Assumed at ~90% for step-down converters routing to 5V logic.**
- **Mission Profile (25 Minutes Total):**
  - **Navigation & Searching: TurtleBot navigates a maze using Pi, OpenCR, LiDAR, and a USB Camera to locate ArUco markers.**
  - **Station A Engagement: Docks at a stationary target, activates a single-flywheel mechanism (RS360 Motor), and fires 3 ping pong balls at 8-second intervals using an SG90 servo feeder.**
  - **Station B Engagement: Docks at a moving target, uses an Ultrasonic sensor to track wall distance, and fires 3 ping pong balls when properly aligned.**

2. Component Power Consumption Breakdown

Component	Operating Voltage (V)	Current Draw (A)	Power (W)	Notes
TurtleBot Base (Pi + OpenCR + LiDAR)	11.1	0.87	9.66	Always Active
1x USB Camera	5.0	0.20	1.11	Always Active (Includes 90% efficiency loss adjustment)
Ultrasonic Sensor	5.0	0.015	0.083	Always Active (Includes 90% efficiency loss adjustment)
1x RS360 Motor (Flywheel)	12.0	0.80	7.20	Active only during target engagement
L298N Motor Driver Loss	12.0	0.30	3.60	~20% thermal/switching loss during engagement

Component	Operating Voltage (V)	Current Draw (A)	Power (W)	Notes
SG90 Servo (Idle)	5.0	0.02	0.11	Holding position during maze navigation
SG90 Servo (Active)	5.0	0.10	0.55	Feeding ping pong balls

3. Phase-Based Energy Calculations

Phase 1: Navigation and Searching (Idle State)

During the majority of the 25-minute mission, the robot navigates the maze and searches for ArUco markers. The flywheel and ball-feeding servo are idle.

Continuous Operations (Base Load)

Your TurtleBot, USB Camera, Ultrasonic Sensor, and the SG90 Servo (sitting in its idle position) are drawing power for the entire 25 minutes.

• **Total Continuous Power**: 9.66 W + 1.11 W + 0.083 W + 0.11 W = 10.963 W

• **Duration**: 25 minutes (0.4167 h)S

• **Energy Consumption:** 10.963 W * 0.4167 h = 4.567 Wh

Phase 2: Target Engagement (Shooting State)

Once docked, the robot activates the RS360 motor (flywheel) and uses the SG90 servo to feed a total of 6 balls across both stations.

- **Station A Timing: 3 balls fired 8 seconds apart. Estimated max flywheel run time = 5.25*3 seconds ~16 seconds**
- **Station B Timing: Tracking wall distance via ultrasonic and firing 3 balls. Estimated max flywheel run time = 120 seconds**
- **Total Flywheel Run Time (t_flywheel): 136 seconds (0.0378 h)**
- **Servo Actuation Time (t_servo): Assuming a standard sweep speed of 0.18s per 90° rotation. 6 balls × 0.18s = 1.08 seconds (0.0003 h)**
- **Shooting Power Overhead (P_shoot): RS360 Motor (7.2 W) + L298N Loss (3.6 W) = 10.8 W**
- **Energy Consumption (Motor): 10.8 W × 0.0378 h = 0.408 Wh**
- **Energy Consumption (Servo Active Difference): (0.55 W - 0.11 W) × 0.0003 h = 0.0001 Wh**

4. Final Power Budget Summary

- **Total Energy Required: 4.567 Wh (Navigation) + 0.408 Wh (Engagement) = 4.976 Wh**
- **Total Battery Capacity: 19.98 Wh**

Conclusion: To determine the endurance of the system, we calculate the number of full 25-minute cycles the battery can support:

`Cycles = 19.98 Wh / 4.976 Wh/cycle ≈ 4.01 cycles`

The system is highly power-efficient. Even accounting for unforeseen electrical inefficiencies and mechanical friction, the 11.1V 1800mAh battery can comfortably sustain the robot for **four complete 25-minute maze iterations** before requiring a recharge.